MediGuardian AI: An Agentic and Service-Oriented Framework for Autonomous Healthcare Navigation

Research Paper Draft — Methodology, Implementation, Outcome, Novelty and Working Model

Abstract

MediGuardian AI is an agentic and service-oriented framework for healthcare navigation. The implemented prototype accepts a natural-language patient request and processes it through an Agent Orchestrator, Intent Agent, healthcare-service mapping, urgency assessment, resource discovery and appointment management. The system is implemented using Spring Boot, Java 21, REST APIs, Spring Data JPA/Hibernate and PostgreSQL, with a lightweight HTML/CSS/JavaScript frontend. The research direction is supported by recent work on agentic healthcare systems and a 2026 multi-agent self-triage framework, while service-oriented healthcare architecture provides the modular integration foundation. The prototype is a software research demonstration and is not a clinical diagnostic or emergency-care system.

1. Introduction

Healthcare navigation often requires a patient to translate a free-text concern into a suitable specialty, healthcare service, provider and next administrative action. Recent research shows that agentic systems can process information, plan and decide, interact and collaborate, use tools and act in healthcare settings. A 2026 Nature Health study demonstrated a multi-agent self-triage system using retrieval, decision and conversation agents. MediGuardian AI focuses on the navigation layer, connecting patient intent with healthcare resources and appointment workflow.

2. Real-Time Base Paper and Related Research

Primary base paper: Y. Liu et al., “A multi-agent framework combining large language models with medical flowcharts for self-triage,” Nature Health, 2026. Published 20 April 2026. The system uses retrieval, decision and conversation agents and reports 84.1% flowchart-retrieval accuracy on 2,000 synthetic opening statements and 99.1% navigation accuracy across 37,200 synthetic patient responses.

Agentic foundation: J. Qiu et al., “LLM-based agentic systems in medicine and healthcare,” Nature Machine Intelligence, 2024. The article describes agentic systems as able to process information, plan and decide, interact and collaborate, use tools and act.

Service-oriented foundation: A. Wright and D. F. Sittig, “SANDS: A Service-Oriented Architecture for Clinical Decision Support in a National Health Information Network,” Journal of Biomedical Informatics, 2008. SANDS demonstrates modular, service-oriented clinical decision support and integration through defined interfaces.

3. Problem Statement

The problem is to design a modular software framework that can transform a natural-language healthcare request into a structured navigation process covering specialty identification, healthcare-service identification, urgency assessment, provider/service discovery and appointment management.

4. Objectives

- Identify healthcare intent from patient text.
- Map intent to a medical specialty and service.
- Assess request urgency using explicit indicators.
- Retrieve matching healthcare resources from a persistent service layer.
- Coordinate components through an Agent Orchestrator.
- Support appointment creation and availability-based confirmation.

• Return a structured navigation response through a REST API and web interface.

5. Methodology Implemented

The implemented methodology follows a modular pipeline. A patient submits a natural-language request through the frontend. The REST API forwards the request to the Agent Orchestrator. The Intent Agent identifies the medical specialty and maps it to a healthcare service. Urgency processing classifies the request. Repository components retrieve healthcare resources from PostgreSQL. Appointment processing evaluates availability and produces a confirmation state. The orchestrator combines these outputs into a single navigation response.

5.1 Working Model

```
Patient Query
↓
HTML/CSS/JavaScript Interface
↓
POST /api/navigate
↓
Agent Orchestrator
↓
Intent Agent → Specialty Identification
↓
Service Identification + Urgency Assessment
↓
Healthcare Resource Repositories
↓
Doctor / Hospital / Service Results
↓
Appointment Management
↓
Structured Navigation Response
```

5.2 Implemented Modules

• **Intent Agent:** identifies specialties from healthcare-related terms and maps them to services.
• **Agent Orchestrator:** coordinates navigation processing and assembles the response.
• **Healthcare services:** represent doctors, hospitals and services as separate backend resources.
• **Persistence:** JPA/Hibernate stores entities in PostgreSQL.
• **Appointment module:** creates appointments and applies the implemented availability rule.
• **REST layer:** exposes navigation and resource operations to the frontend.

6. Implementation

The prototype uses Java 21 and Spring Boot 3.5.5. Spring Web provides REST APIs; Spring Data JPA and Hibernate provide persistence; PostgreSQL provides the database; and HTML/CSS/JavaScript provide the client interface. The navigation endpoint is POST /api/navigate and accepts a patient identifier and natural-language query. The response contains specialization, service, urgency, matching resources and the next action.

7. Outcome

The prototype demonstrates an end-to-end navigation workflow. During functional testing, a natural-language orthopedic request was mapped to the Orthopedician specialty and Orthopedic Consultation service, and matching provider information was returned. Appointment tests also demonstrated confirmation for an available time and non-confirmation for an out-of-availability test case. These are software-functional outcomes, not clinical validation results.

8. Novelty / Research Contribution

The novelty should not be claimed as the invention of multi-agent healthcare AI, because recent research already demonstrates multi-agent self-triage and agentic healthcare systems. The

proposed contribution is the integration of patient-intent analysis, specialty/service mapping, urgency assessment, healthcare-resource discovery and appointment management into one service-oriented navigation workflow. The architecture combines agentic coordination with modular healthcare services and extends navigation beyond a recommendation toward an operational appointment step. This is conceptually related to the modularity and integrability goals of service-oriented clinical decision support.

9. Discussion and Limitations

The architecture is modular and allows individual agents or services to be extended independently. However, the current prototype uses primarily rule-based intent and urgency logic rather than a clinically validated machine-learning or large-language-model classifier. The provider data and availability rules are prototype data, and the system has not undergone clinical validation. Therefore, the current results demonstrate technical feasibility rather than medical accuracy or safety.

10. Future Work

Future work can include validated NLP/LLM intent recognition, structured healthcare terminology, real-time provider calendars, authentication and audit logging, privacy controls, clinically reviewed evaluation datasets, latency and usability studies, and interoperability with standardized healthcare data services.

11. Conclusion

MediGuardian AI demonstrates a working agentic and service-oriented software framework for autonomous healthcare navigation. It connects patient queries with specialty identification, healthcare-service mapping, urgency assessment, resource discovery and appointment management using Spring Boot REST services and PostgreSQL. Its contribution is best positioned as an integrated navigation architecture built on established agentic and service-oriented concepts, rather than as a claim that those underlying technologies are themselves new.

References

[1] Liu, Y. et al., “A multi-agent framework combining large language models with medical flowcharts for self-triage,” Nature Health, vol. 1, pp. 840–849, 2026. DOI: 10.1038/s44360-026-00112-2.

[2] Qiu, J. et al., “LLM-based agentic systems in medicine and healthcare,” Nature Machine Intelligence, vol. 6, pp. 1418–1420, 2024. DOI: 10.1038/s42256-024-00944-1.

[3] Wright, A. and Sittig, D. F., “SANDS: A Service-Oriented Architecture for Clinical Decision Support in a National Health Information Network,” Journal of Biomedical Informatics, vol. 41, no. 6, pp. 962–981, 2008. DOI: 10.1016/j.jbi.2008.03.001.